

Hung Nguyen

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RESEARCH INTERESTS

Frequency-domain learning and compression for computer vision, e.g., via Fourier, cosine, and wavelet transforms.
Perceptual quality assessment of compressed visual representations.

EDUCATION

- **University of California San Diego** Sep 2026 - Present
Doctor of Philosophy in Electrical Engineering (Signal and Image Processing) San Diego, USA
- **University of California San Diego** Sep 2023 - Jun 2025
Master of Science in Electrical Engineering (Signal and Image Processing) San Diego, USA
- **Ho Chi Minh City University of Technology** Aug 2017 - Jun 2022
Bachelor of Engineering in Mechatronics Engineering Ho Chi Minh, Vietnam

RESEARCH EXPERIENCE

- **University of California San Diego**  Sep 2024 - Present
Graduate Student Researcher San Diego, USA
 - Developed frequency-domain methods for efficient Gaussian Splatting, with focus on learnable wavelet transforms.
 - Published first-author research at ICCV, VCIP, ICASSP, and EUSIPCO conferences, co-advised by Distinguished Profs. [Truong Nguyen](#) and [Pamela Cosman](#) at the [Video Processing Lab](#).

WORK EXPERIENCE

- **VinMotion**  Dec 2025 - May 2026
Senior AI Engineer Hanoi, Vietnam
 - Researched multi-modal algorithms for humanoid robots, particularly speech-conditioned grasp detection.
- **WorldQuant**  Jan 2023 - Jul 2023
Quantitative Researcher Intern Ho Chi Minh, Vietnam
 - Developed CNN models for stock price prediction and evaluated their performance on historical market data.

HONORS AND AWARDS

- **ECE Department Fellowship** Feb 2026
UC San Diego 
- **Vingroup Scholarship for Overseas Study for Master's and Doctoral Degrees** Jun 2023
Vingroup JSC 
- **Panasonic Asia Scholarship for Undergraduate Students** Apr 2022
Panasonic Vietnam 

FIRST-AUTHOR PUBLICATIONS & PREPRINTS

- [C.1] **Hung Nguyen**, An Le, Runfa Li, Truong Nguyen. [From Coarse to Fine: Learnable Discrete Wavelet Transform for Efficient 3D Gaussian Splatting](#). ICCV workshop 2025.
- [C.2] **Hung Nguyen**, Runfa Li, An Le, Truong Nguyen. [DWTGS: Rethinking Frequency Regularization for Sparse-view 3D Gaussian Splatting](#). VCIP 2025.
- [C.3] **Hung Nguyen**, Runfa Li, An Le, Truong Nguyen. [WaveletGaussian: Wavelet-domain Diffusion for Sparse-view 3D Gaussian Object Reconstruction](#). ICASSP 2026.
- [C.4] **Hung Nguyen**, An Le, Truong Nguyen. [Learnable Multi-level Discrete Wavelet Transforms for 3D Gaussian Splatting Frequency Modulation](#). EUSIPCO 2026.
- [C.5] **Hung Nguyen***, Kim Nhat Minh Nguyen*, Van Duc Vu, Van-Danh Le, Hoang Huy Le, Dinh Tuan Nguyen, Pham Tuyen Le, Van-Truong Nguyen, Quan Nguyen. [Speech2Grasp: Data-Efficient Transfer of Text Conditioned Grasp Detection to Speech in Humanoid Robots](#). arXiv preprint.

SKILLS

- **Programming Languages:** Python, C/C++, MATLAB
- **DL/ML Packages:** PyTorch, TensorFlow, Keras, JAX, standard image processing libraries (e.g., OpenCV, NumPy)
- **DL/ML Methods:** CNNs, Vision Transformers, Diffusion Models, NeRF, 3DGS, etc.